

1a) Given:

Internal combustion engine, moderate load.

No. 40 roller chain, single strand.

$$P = 6 \text{ kW}$$

$$\eta_{\text{engine}} = 2750 \text{ rpm}$$

$$\eta_{\text{machine}} = 800 \pm 10\%$$

From Table B-2:

Service factor = 1.4

Internal combustion engine with mechanical drive

$$\text{Design power per strand} = \frac{1.4 \times 6}{1} = 8.4 \text{ kW}$$

$$\begin{aligned} \eta_c = \eta_B &= \frac{D_J \eta_J}{D_B} \\ &= \frac{118(2750)}{180} \\ &= \frac{16225}{9} \approx 1803 \text{ rpm} \end{aligned}$$

From table B-4:

$$N_c = 21$$

$$1a) \eta_E = \frac{790 + 810}{2}$$

$$= 800$$

$$N_E = N_C \left(\frac{\frac{16225}{9}}{800} \right)$$

$$= 21 \left(\frac{649}{288} \right)$$

$$= 47.32291667$$

$$\approx 47$$

Taking $N_E = 47$, checking η_E :

$$\eta_E = \frac{21 \left(\frac{16225}{9} \right)}{47}$$

$$= 805.4964539$$

$$\approx 805 \text{ rpm (within range)}$$

$\therefore$ Shaft L rotates at 805 rpm and rotates anti-clockwise.

1b) From Table B-1:

For No. 40 roller chain:

$$P = 12.70 \text{ mm}$$

$$D_E = \frac{P}{\sin\left(\frac{180}{47}\right)}$$

$$= \frac{12.7}{\sin\left(\frac{180}{47}\right)}$$

$$= 190.1407279$$

$$\approx 190 \text{ mm}$$

$$P_E = T_E \omega_E$$

$$T_E = \frac{P_E}{\omega_E} = \frac{6 \times 10^3}{\frac{2\pi}{60} \times 805}$$
$$= 71.13101397 \text{ Nm}$$
$$\approx 71 \text{ Nm}$$

$$T_E = F_E \left(\frac{D_E}{2} \right)$$

$$F_E = \frac{2T_E}{D_E} = \frac{2(71)}{190 \times 10^{-3}}$$

$$= 748.1933486 \text{ N}$$

$$\approx 748.2 \text{ N}$$

$$\therefore \bar{F}_1 = 748.2 \text{ N}$$
$$F_2 = 0 \text{ N (slack)}$$